1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlighted that lncRNA-mediated control of TGF-β signaling (TGFB2■AS1), and a TP53/CTNNB1–miR■183 stemness axis, converge to produce divergent prognoses in NMIBC — specifically, why tumors with apparently similar stromal or microenvironmental features show opposite clinical outcomes. 2) Tier 1 paths used: I built on the TGFB2■AS1 path (core lncRNA node modulating TGF■β/EMT and stromal activation) and the TP53/CTNNB1→miR■183→stemness path (CTNNB1 as a driver of cell■intrinsic stemness/EMT programs). These two provide a clear mechanistic tension: stromal/TGF■β regulation versus tumor■intrinsic stemness. 3) Tier 2 support: JAK3 (immune/JAK–STAT signaling), HSPE1 (proteostasis/stress adaptation), OIP5 (stemness/cell cycle), and the MIS/hormonal axis offer plausible modifiers that can shape stromal activation, immune contexture, and tumor cell resilience, reinforcing the proposed mechanism by which stromal signals are either restrained or overcome. 4) Endpoint choice: Overall survival (OS) was chosen as the single primary endpoint because it is the most definitive clinical outcome available across cohorts and best captures the net effect of microenvironment–tumor interactions on patient prognosis. 5) Deliberate exclusions: I excluded broad, ill■defined 'hub gene' labels and dispersed Tier■3 candidates (YWHAE, FUT3) from the core claim to preserve mechanistic clarity and to respect the 1 imaging + ≤2 anchor gene constraint; those broader network hubs and exploratory genes are retained for downstream secondary analyses rather than the primary hypothesis?

Working Hypothesis

In TSR-high tumors, the prognostic impact is determined by the relative activity of tumor cell–intrinsic CTNNB1-driven stemness and stromal TGFB2-AS1–modulated TGF-β signaling. TGFB2-AS1 dampens TGF-β–driven stromal activation and EMT, while CTNNB1 promotes EMT and stemness; theNet effect on overall survival (OS) depends on which program dominates. Specifically, within TSR-high contexts: (a) TGFB2-AS1-high and CTNNB1-low tumors will exhibit the best OS due to restrained stromal remodeling and limited stemness; (b) TGFB2-AS1-low and CTNNB1-high tumors will exhibit the worst OS due to unrestrained TGF-β–driven stroma and strong tumor cell stemness; (c) TGFB2-AS1-high/CTNNB1-high tumors will show intermediate OS where CTNNB1-driven stemness partially offsets TGFB2-AS1-mediated restraint. In TSR-low tumors, the influence of TGFB2-AS1 and CTNNB1 is diminished, yielding comparatively intermediate OS with less pronounced interactions. Tier 2 modifiers (TP53, JAK3, HSPE1, OIP5) will modulate these effects by shaping immune contexture, proteostasis/stress responses, and cell cycle/stemness, refining risk stratification across TSR and anchor-gene strata. The primary endpoint is OS in TCGA cohorts, with secondary endpoints including progression-free survival and metastasis-free survival and supportive analyses using TIL/TLS density and immune signatures to link microenvironment context with genomic/epigenomic states.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TGFB2-AS1, CTNNB1
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB2-AS1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

CTNNB1: CTNNB1–miR■183 stemness axis, converge to produce divergent prognoses in NMIBC — specifically, why tumors with apparently similar stromal or microenvironmental features show opposite clinical outcomes.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	JAK3	Supporting modifier from knowledge graph
Secondary	HSPE1	Supporting modifier from knowledge graph
Secondary	OIP5	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In silico: analyze TCGA-BRCA and other solid tumor datasets with available imaging-derived TSR or feasible proxies; compute TGFB2-AS1 and CTNNB1 activity, adjust for confounders; test interaction terms in Cox models and perform stratified analyses by TSR.

Prediction 2: In vitro: manipulate TGFB2-AS1 and CTNNB1 in breast cancer cell lines; assess TGF-β signaling (SMAD2/

Prediction 3: , EMT markers (E-cadherin, N-cadherin), stemness (CD44, ALDH

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- JAK3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.